

Do-While Loops In Java

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What We'll Learn

By the end of this lesson, students can:

- Explain how a do-while loop repeats statements.
- Identify the loop body and condition.
- Compare while and do-while loops.
- Create Java programs that execute a task at least once.
- Use a do-while loop with user input.

Before We Start

Consider a program that asks a user to enter a number.

The program must ask for the number at least once before it can determine whether the input is valid.

How can we create a loop that always executes its statements once before checking its condition?

Let's Understand

A do-while loop repeats statements while a condition remains true.

Unlike a while loop, the do-while loop checks its condition after executing the loop body.

This means the statements inside a do-while loop always execute at least once.

Basic Syntax

```
do {  
    // Statements to repeat  
} while (condition);
```

The semicolon after the while condition is required.

How A Do-While Loop Works

1. Java executes the statements inside the do block.
2. Java checks the condition.
3. If the condition is true, the loop body executes again.
4. The loop continues while the condition remains true.
5. The loop stops when the condition becomes false.

Counter-Controlled Do-While Loop

```
int number = 1;  
  
do {  
    System.out.println(number);  
    number++;  
} while (number <= 5);
```

Output:

```
1  
2  
3  
4  
5
```

This loop contains three important parts:

- Initialization: `int number = 1;`
- Update: `number++;`
- Condition: `number <= 5`

Tracing The Loop

Value displayed	Updated value	Is number <= 5?	Result
1	2	true	Repeat
2	3	true	Repeat
3	4	true	Repeat
4	5	true	Repeat
5	6	false	Stop

Do-While Loop Executes At Least Once

Study the following example:

```
int number = 10;

do {
    System.out.println(number);
} while (number < 5);
```

Output:

```
10
```

The condition `number < 5` is false, but the value 10 is still displayed because the condition is checked after the loop body executes.

While And Do-While Comparison

Using a while loop:

```
int number = 10;

while (number < 5) {
    System.out.println(number);
}
```

This loop executes zero times because the condition is checked first.

Using a do-while loop:

```
int number = 10;

do {
```

```
    System.out.println(number);  
} while (number < 5);
```

This loop executes once because the condition is checked after the loop body.

Loop	Condition Checked	Minimum Repetitions
while	Before the loop body	0
do-while	After the loop body	1

Preventing Infinite Loops

An infinite loop occurs when the condition never becomes false.

```
int number = 1;  
  
do {  
    System.out.println(number);  
} while (number <= 5);
```

The value of number never changes, so the condition always remains true.

The loop variable must be updated:

```
int number = 1;  
  
do {  
    System.out.println(number);  
    number++;  
} while (number <= 5);
```

Let's Practice

Practice 1: Trace The Loop

Study the following program:

```
public class LoopTracing {
    public static void main(String[] args) {
        int number = 2;

        do {
            System.out.println(number);
            number = number + 2;
        } while (number <= 10);
    }
}
```

Answer the following questions:

1. What is the initial value of number?
2. What numbers will the program display?
3. How does the value of number change?
4. What condition controls the loop?
5. What is the value of number when the loop stops?

Practice 2: Complete The Loop

Complete the missing parts so the program displays the numbers from 1 to 5.

```
public class CompleteLoop {
    public static void main(String[] args) {
        int count = ____;

        ____ {
            System.out.println(count);
            count____;
        } ____ (count <= 5);
    }
}
```

Practice 3: Compare The Loops

Study the two programs.

Program A:

```
int number = 8;

while (number < 5) {
```

```
    System.out.println(number);  
}
```

Program B:

```
int number = 8;  
  
do {  
    System.out.println(number);  
} while (number < 5);
```

Answer the following:

1. How many times will Program A display the number?
2. How many times will Program B display the number?
3. Why do the two programs produce different results?

Practice 4: Find And Correct The Errors

The following program contains errors:

```
public class FixTheLoop {  
    public static void main(String[] args) {  
        int number = 1;  
  
        do {  
            System.out.println(number);  
            number++;  
        } while (number <= 5)  
    }  
}
```

Identify and correct the error in the program.

Practice 5: Create A Do-While Program

Create a Java program that:

1. Asks the user to enter a number from 1 to 10.
2. Uses a do-while loop.
3. Repeats the question when the number is outside the accepted range.
4. Stops asking when the user enters a valid number.
5. Displays the accepted number.

Resources / References

- [Java Development Kit](#)
- [Visual Studio Code](#)
- [Java do-while loop documentation](#)
- [Java Scanner class](#)
- [Java switch statement documentation](#)